Understand AI Better Than Your Students

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Setting the Stage

Common Perceptions of AI

> “AI can think like a human, and will replace humans in all kinds of fields within the next few years.”

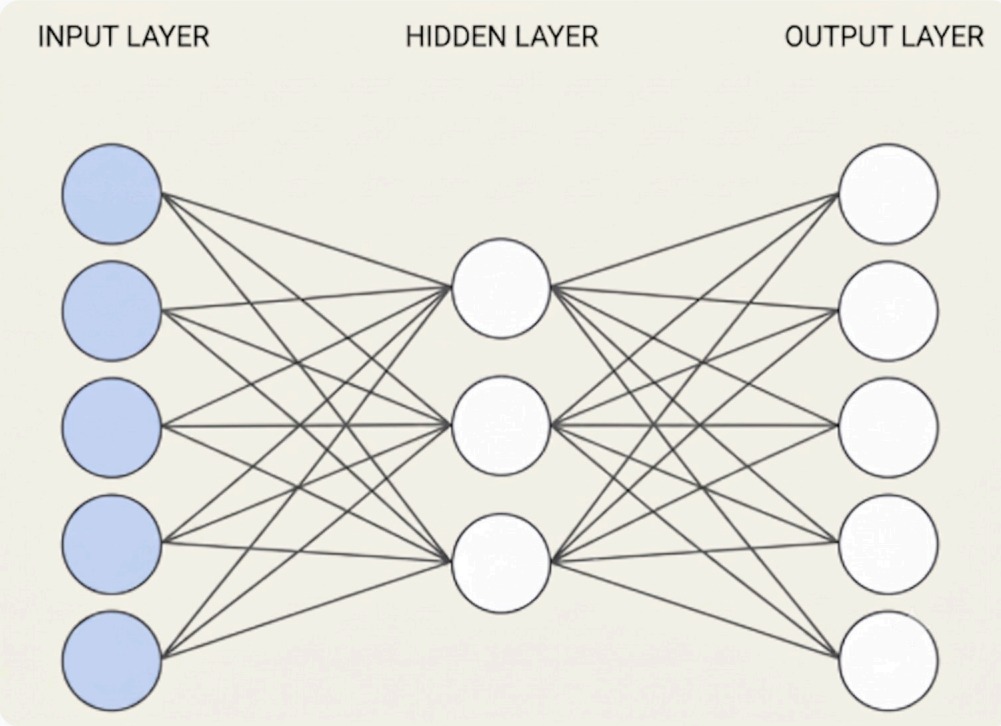
> “AI just repeats back things they found on the internet; they’re basically fancy search engines.”

> “AI constantly makes mistakes, and can’t be relied on for important tasks.”

The Complicated Truth

Understanding the actual systems helps you understand:

- How AI works
- Where it excels
- Where it struggles



Our Focus Today

Not

“How do I use AI?”

But

“How does AI work?”

Questions for Today

1. **What *are* LLMs?**
2. Why are they **good at some things** (coding) but **bad at others** (creative writing)?
3. In what ways **are humans involved**? In what ways **do models "learn on their own"**?

“Human LLM” Activity

Activity Instructions

1. In the top-left corner of your postcard, write one word.
2. Pass it along (snaking through the room).
3. With the postcard you receive, add one more word.
4. Repeat.

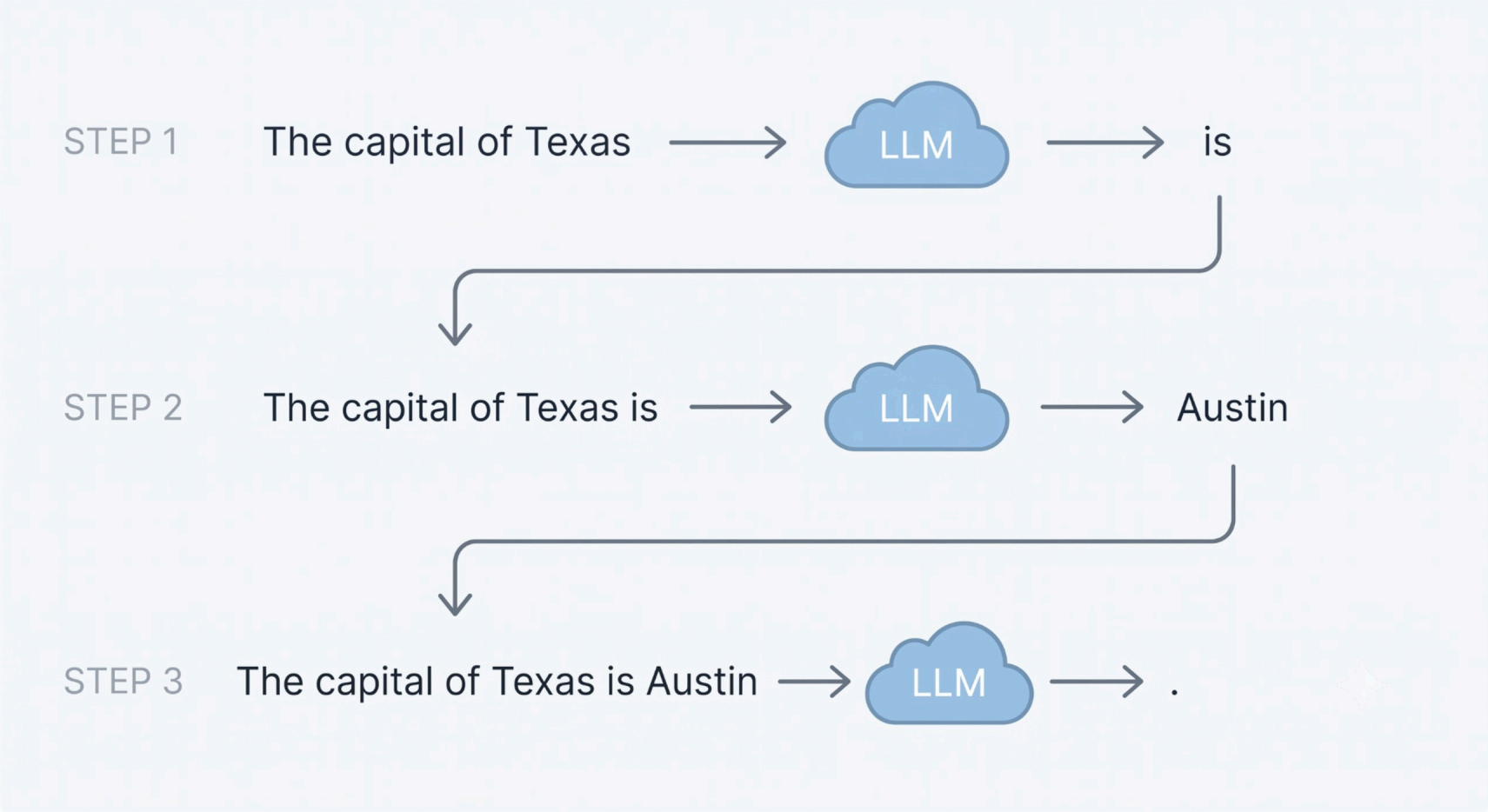
Reflection

What you just did is called **autoregressive generation**.

You generated new words using the previous words as input.

This is how LLMs generate text.

Autoregressive Generation



But what *are* LLMs?